Build a Personalized Online Course Recommender System with Machine Learning

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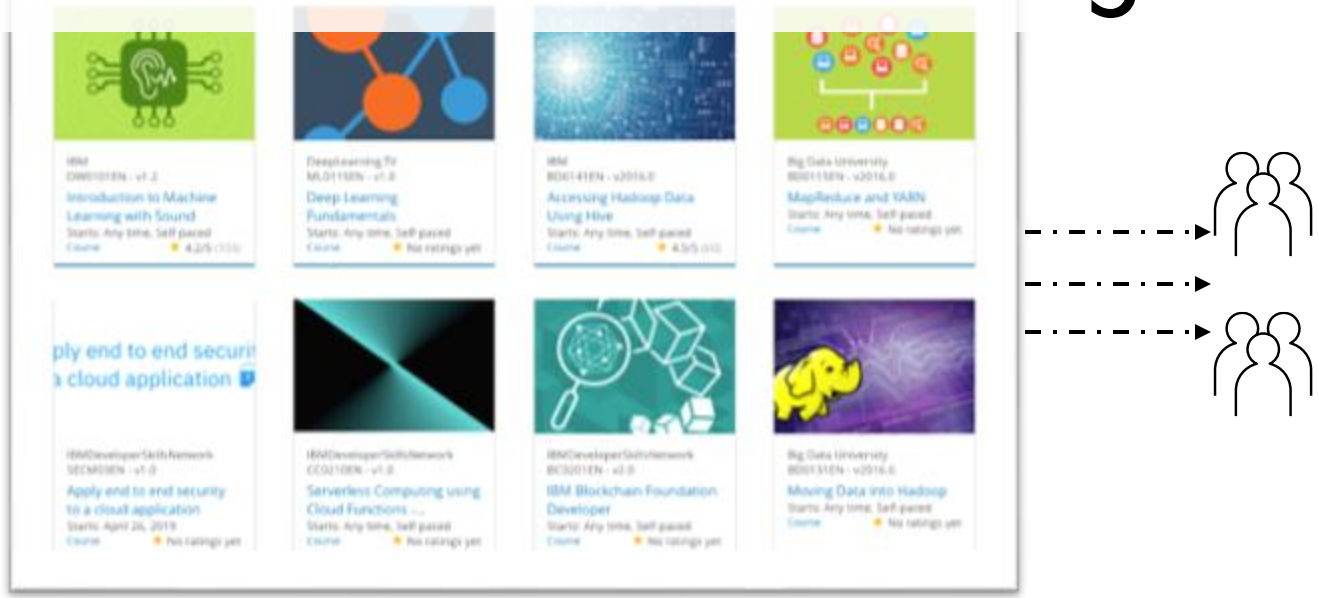


Outline

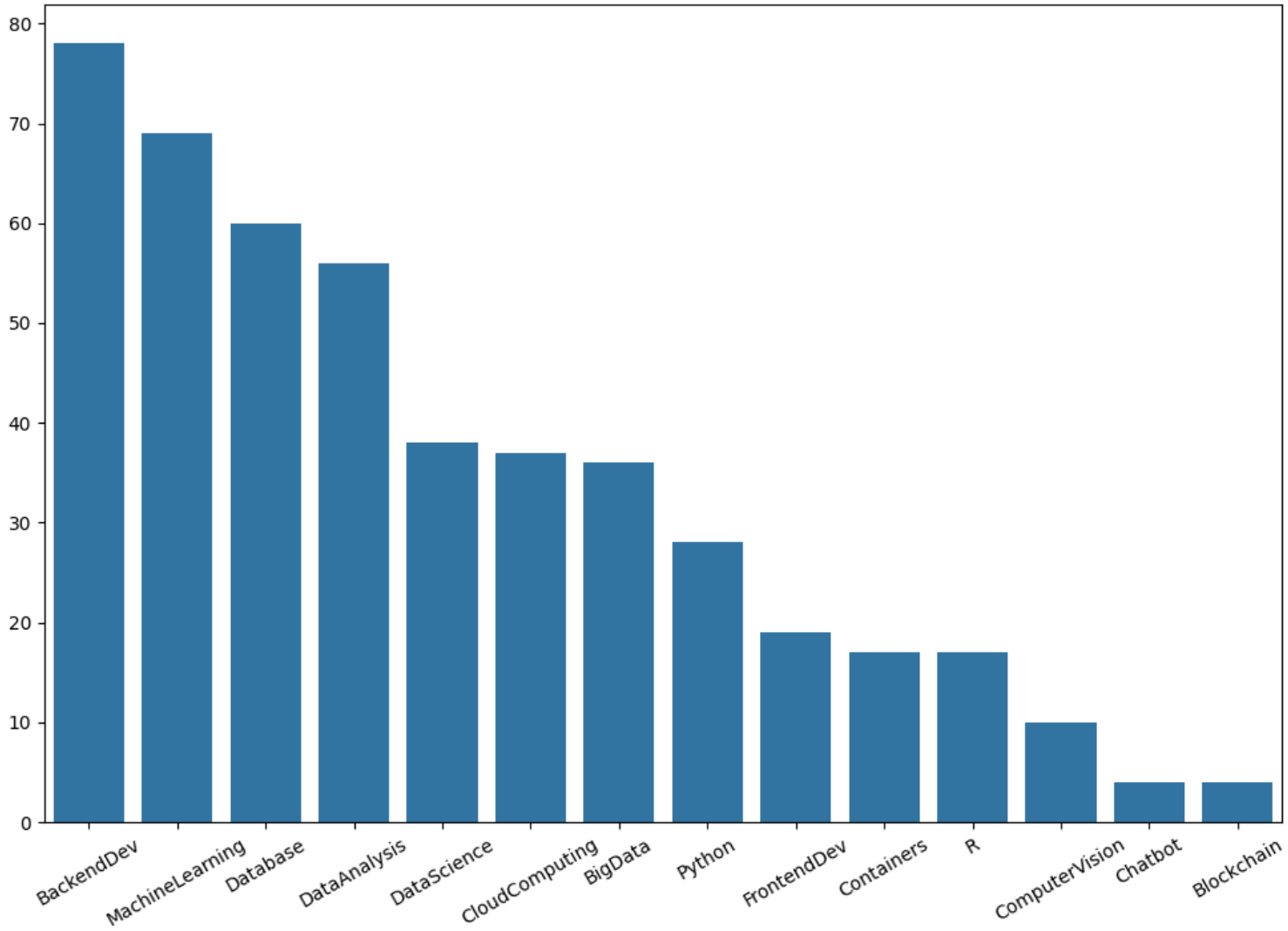
- Introduction and Background
- Exploratory Data Analysis
- Content-based Recommender System using Unsupervised Learning
- Collaborative-filtering based Recommender System using Supervised learning
- Conclusion
- Appendix

Introduction

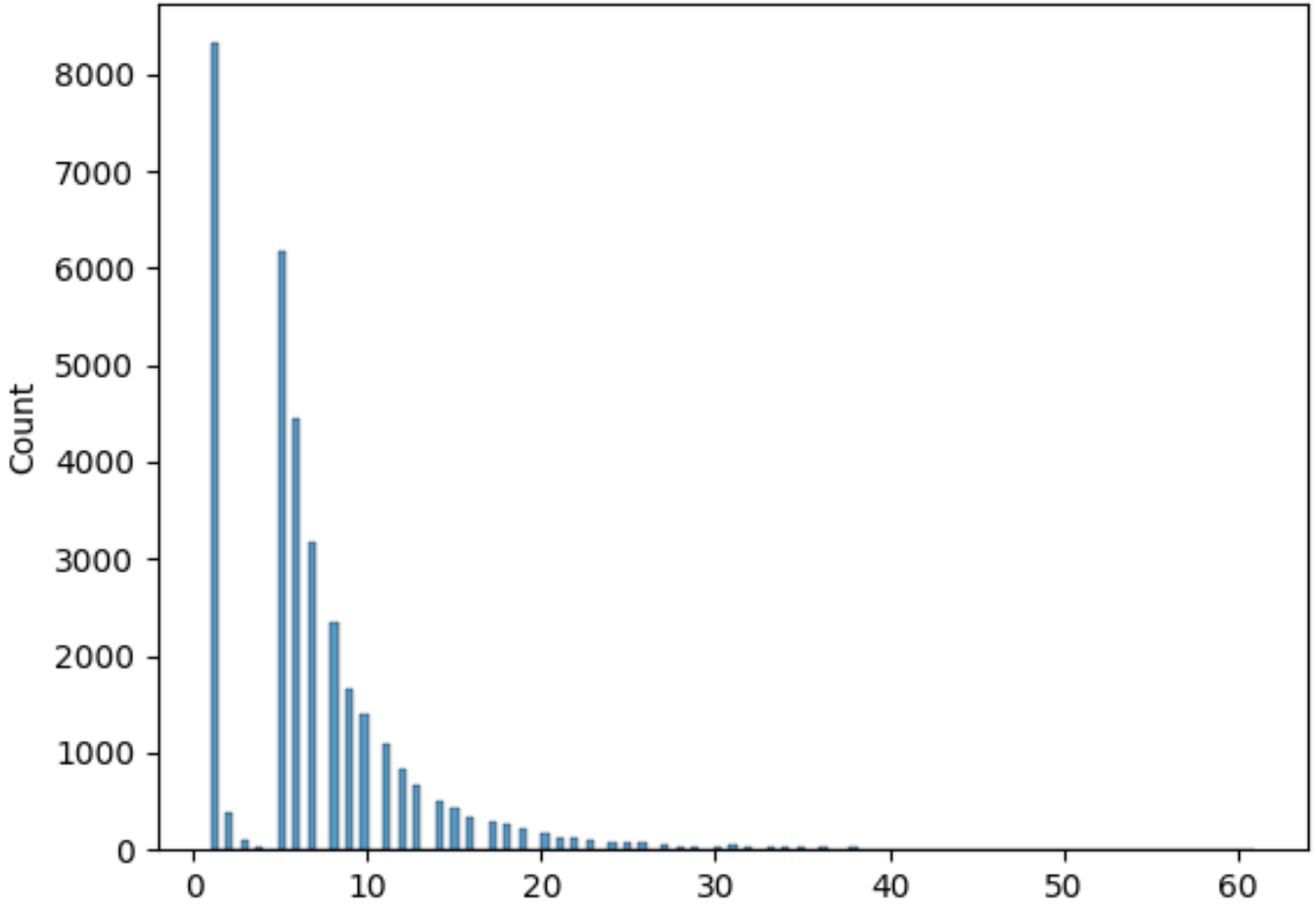
- Online learning platforms such as Coursera offer a plethora of courses
  - Virtually over any branch of knowledge
  - Useful for any kind of job/activity
  - World-class learning for individuals and organizations
- Challenge – Recommend the most adequate courses to millions of learners
  - Based on their goals, aspirations, interests, areas of improvement
  - This project offers Machine Learning solutions for a Personalized Online Course Recommender System
    - Unsupervised methods – User Profile-based, Course Similarity-based, Clustering-based
    - Supervised methods – KNN, NMF, NN Embeddings

Exploratory Data Analysis

Course counts per genre



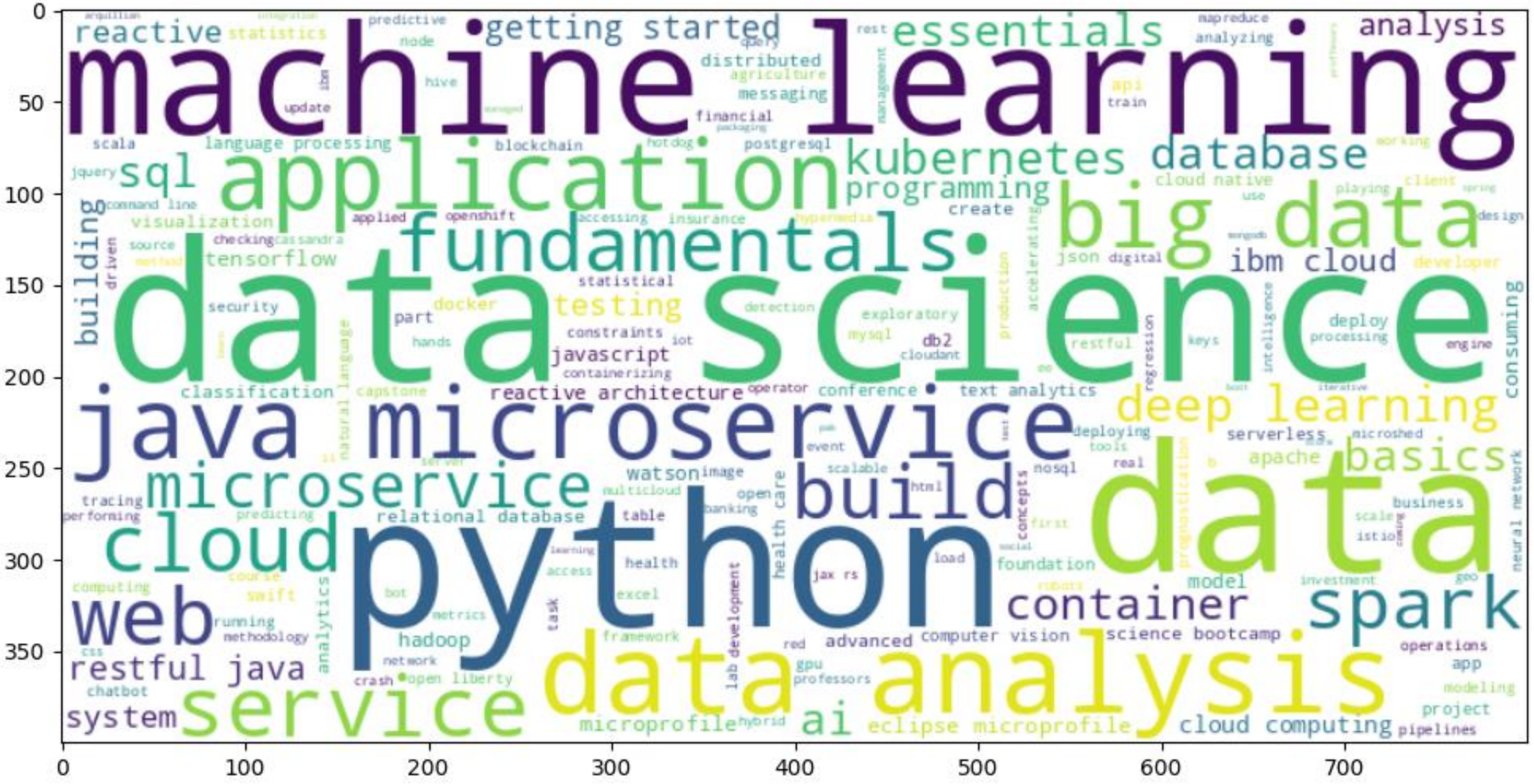
Course enrollment distribution



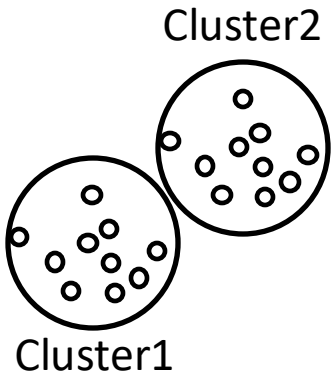
20 most popular courses

	Title	Enrollments
0	python for data science	14936
1	introduction to data science	14477
2	big data 101	13291
3	hadoop 101	10599
4	data analysis with python	8303
5	data science methodology	7719
6	machine learning with python	7644
7	spark fundamentals i	7551
8	data science hands on with open source tools	7199
9	blockchain essentials	6719
10	data visualization with python	6709
11	deep learning 101	6323
12	build your own chatbot	5512
13	r for data science	5237
14	statistics 101	5015
15	introduction to cloud	4983
16	docker essentials a developer introduction	4480
17	sql and relational databases 101	3697
18	mapreduce and yarn	3670
19	data privacy fundamentals	3624

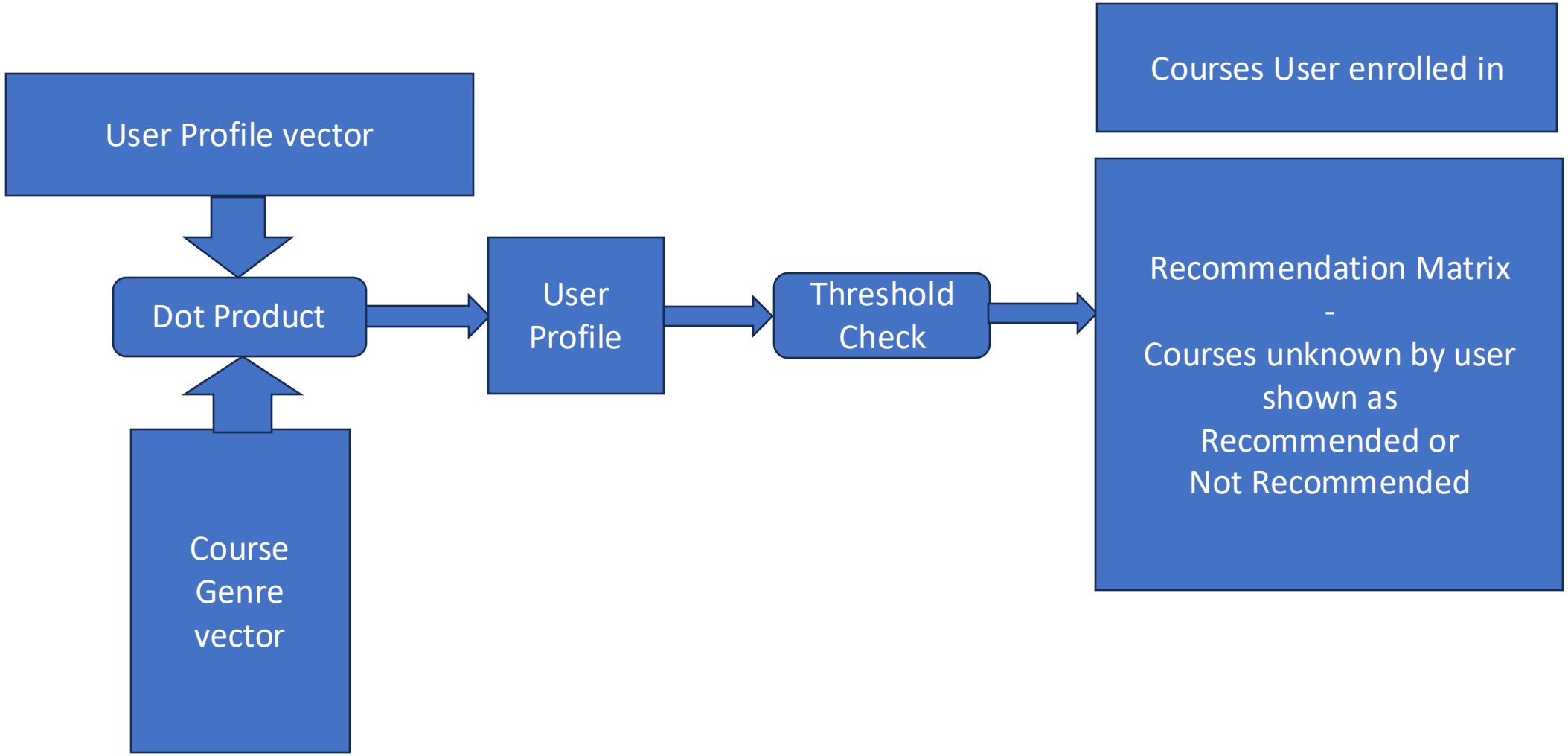
Word cloud of course titles



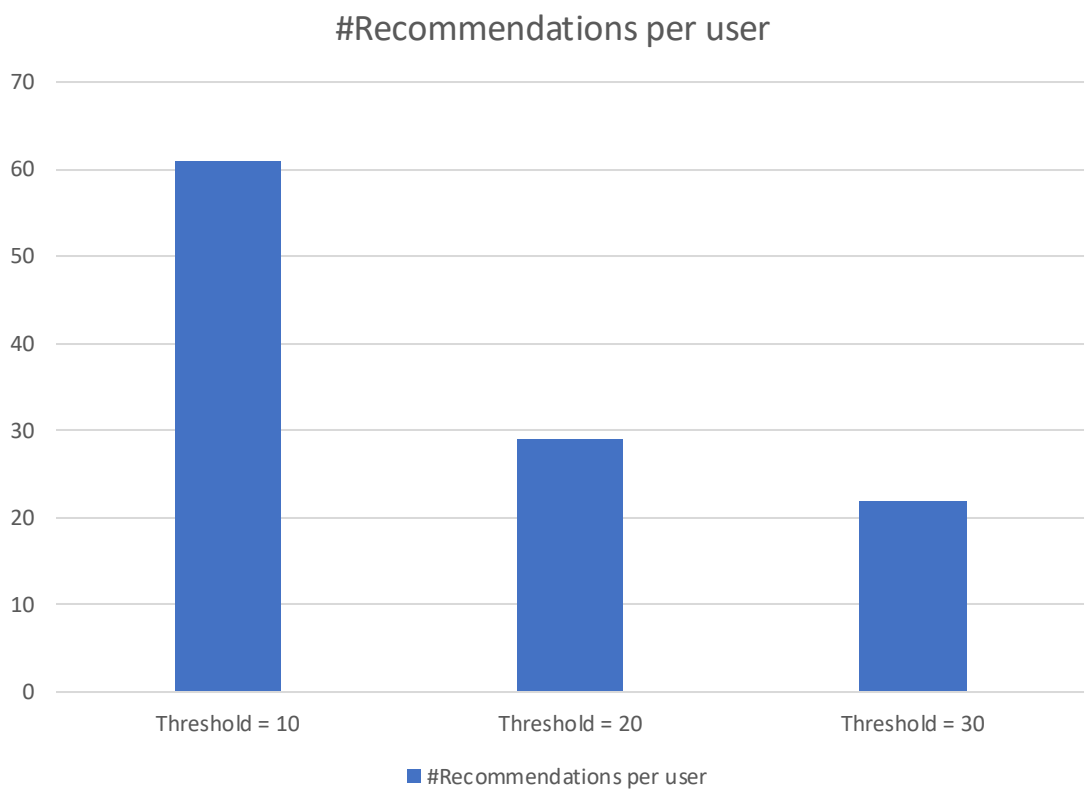
Content-based Recommender System using Unsupervised Learning



Flowchart of content-based recommender system using user profile and course genres



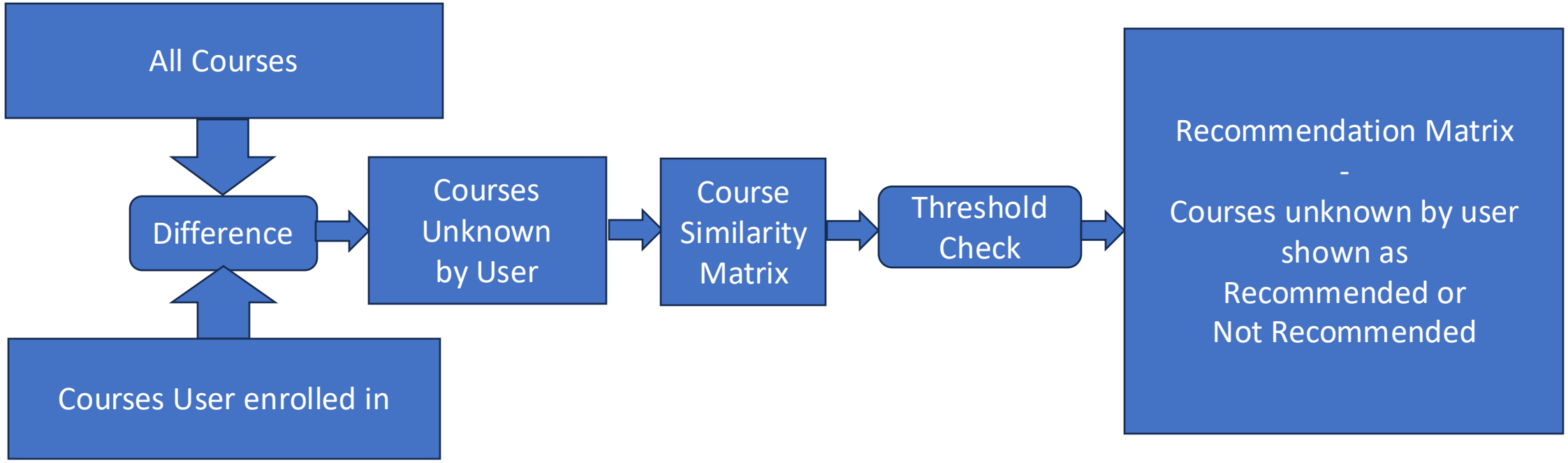
Evaluation results of user profile-based recommender system



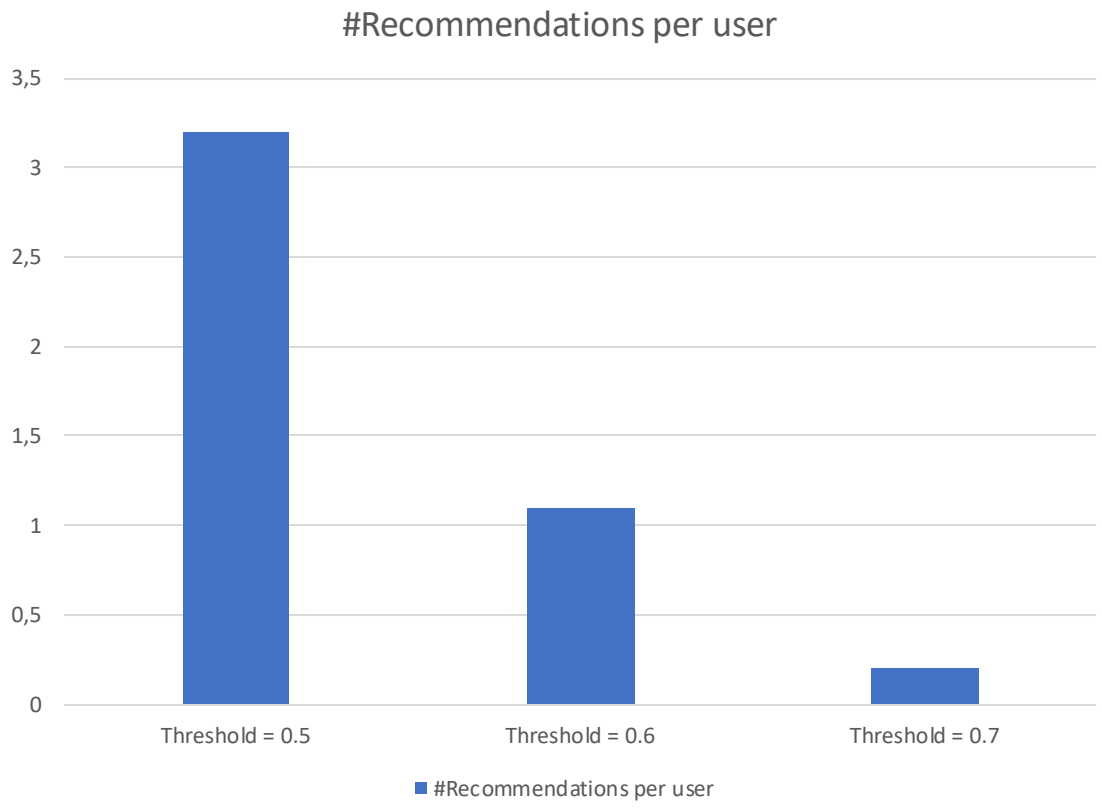
Top-10 recommendations
Threshold = 10

Title	count
introduction to data science in python	28696
accelerating deep learning with gpu	25395
applied machine learning in python	24444
data analysis using python	19150
text analytics at scale	17390
machine learning with python	16451
data science in insurance basic statistical analysis	15644
exploratory data analysis for machine learning	15384
sql for data science	15062
sql for data science capstone project	15062

Flowchart of content-based recommender system using course similarity



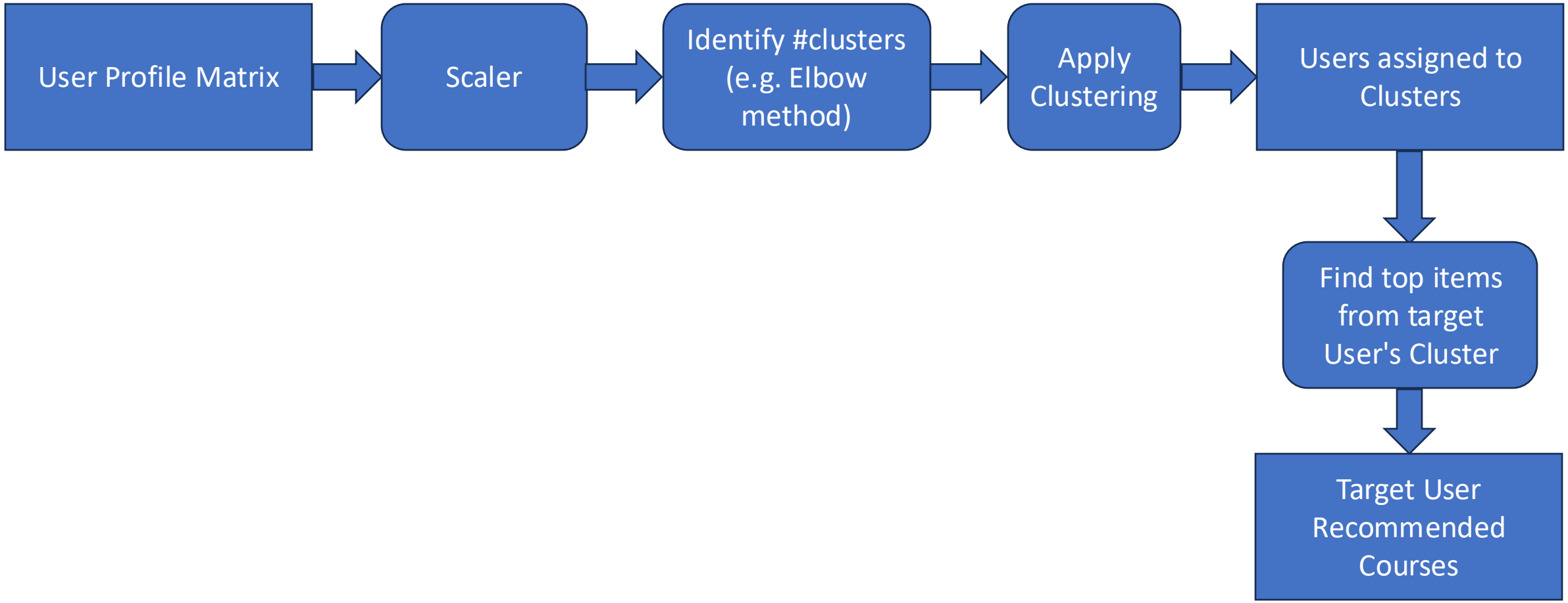
Evaluation results of course similarity-based recommender system



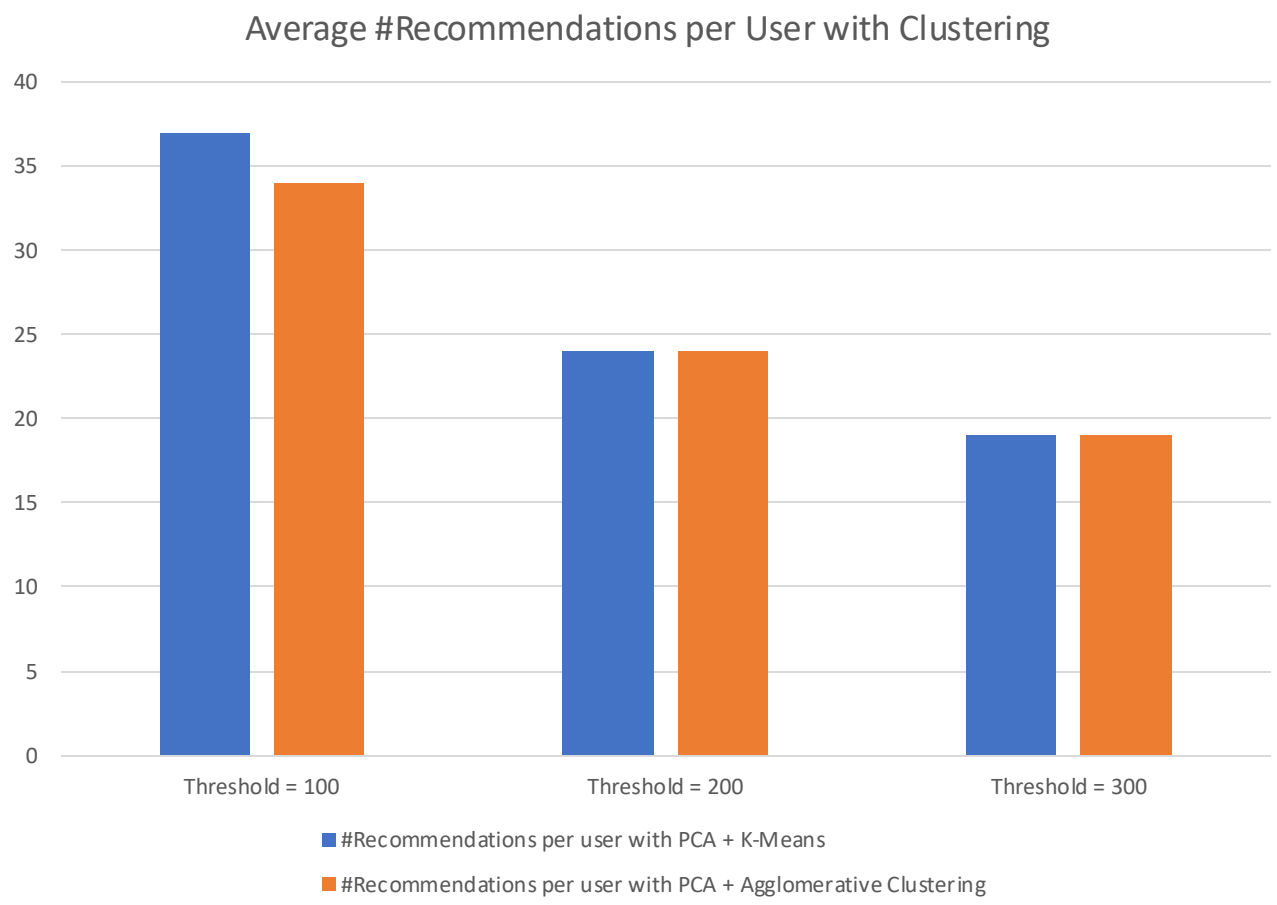
Top-10 recommendations
Threshold = 0.5

	Title	#Recommendations
0	data science bootcamp with python	8841
1	introduction to data science in python	7427
3	introduction to data analytics	6360
4	data science with open data	4900
5	big data modeling and management systems	3423
6	data analysis using r 101	2812
7	data analysis using python	2626
9	data science fundamentals for data analysts	2540
10	process data from dirty to clean	2508
11	\nsql for data science	2442

Flowchart of clustering-based recommender system



Evaluation results of clustering-based recommender system

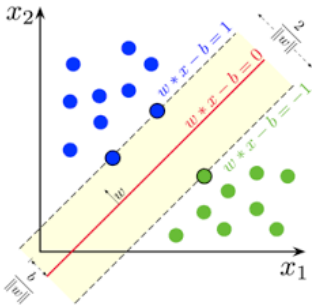


Top-10 recommendations with PCA + K-Means

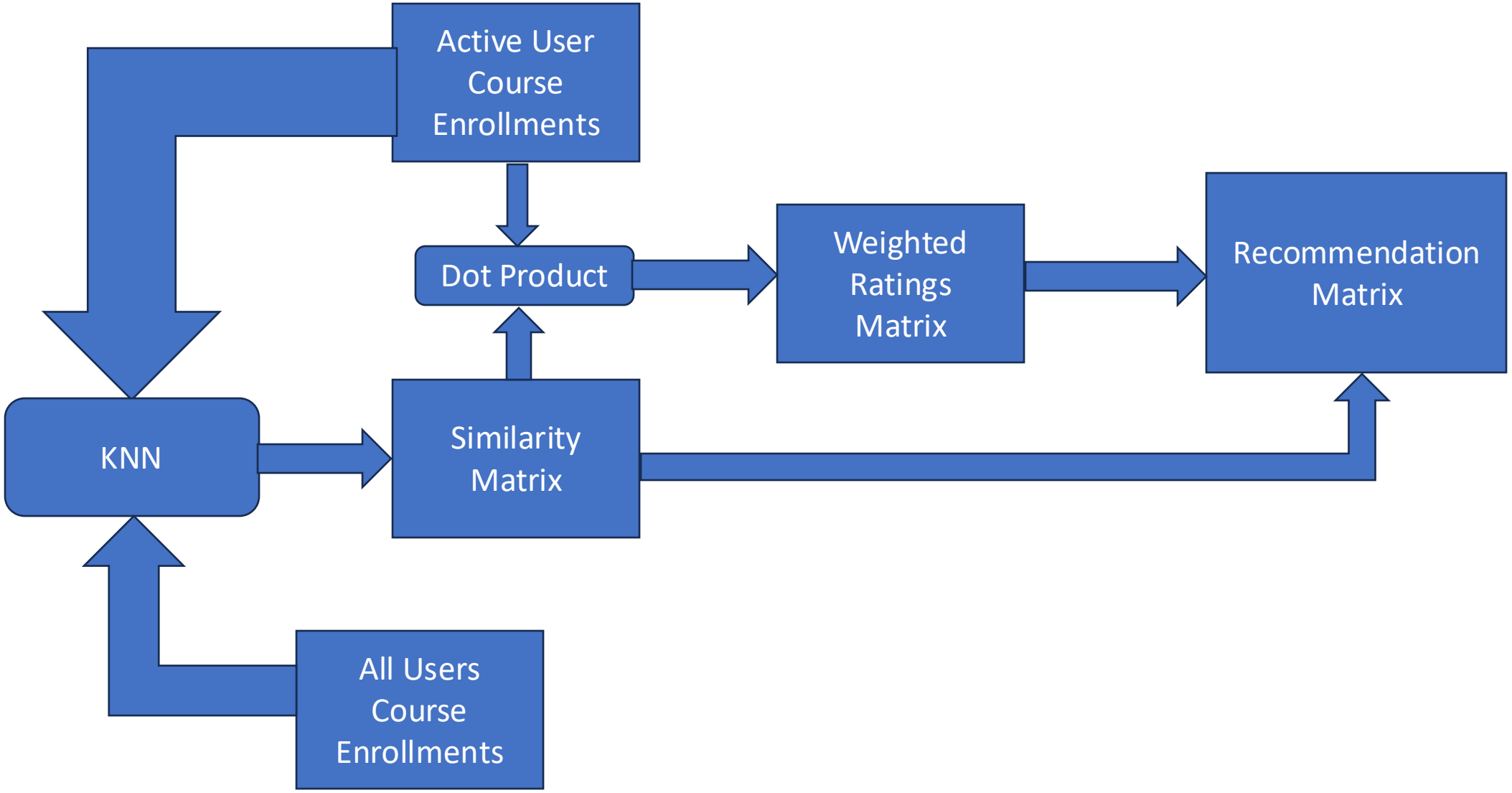
Threshold = 100

Recommendations	count
DS0321EN	956
SC0101EN	927
CL0101EN	927
WA0101EN	911
DB0101EN	901
CO0101EN	896
DS0301EN	892
BD0115EN	885
CC0101EN	871
ML0101EN	854

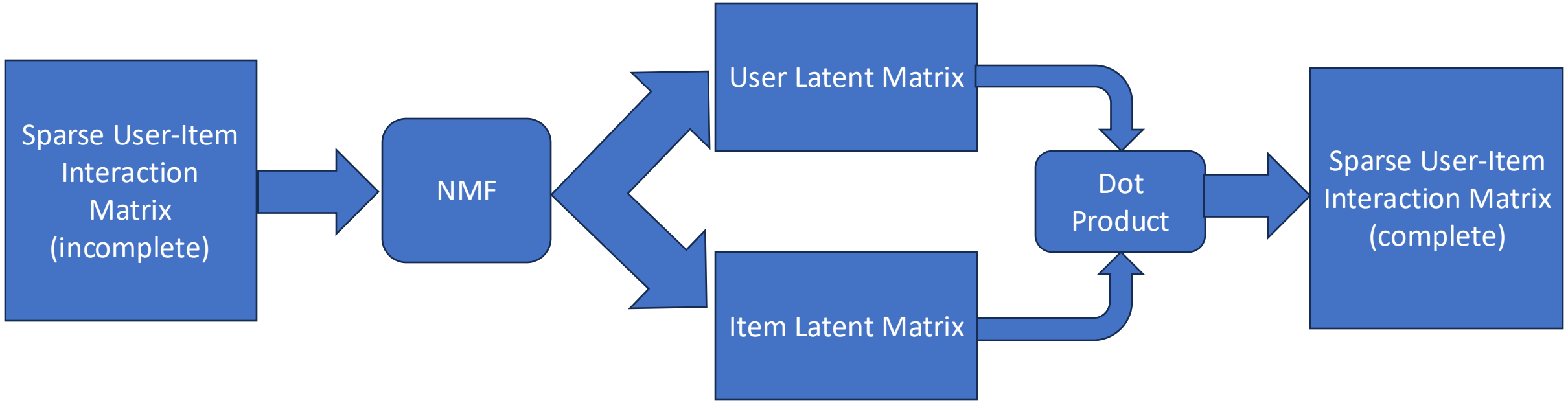
Collaborative-filtering Recommender System using Supervised Learning



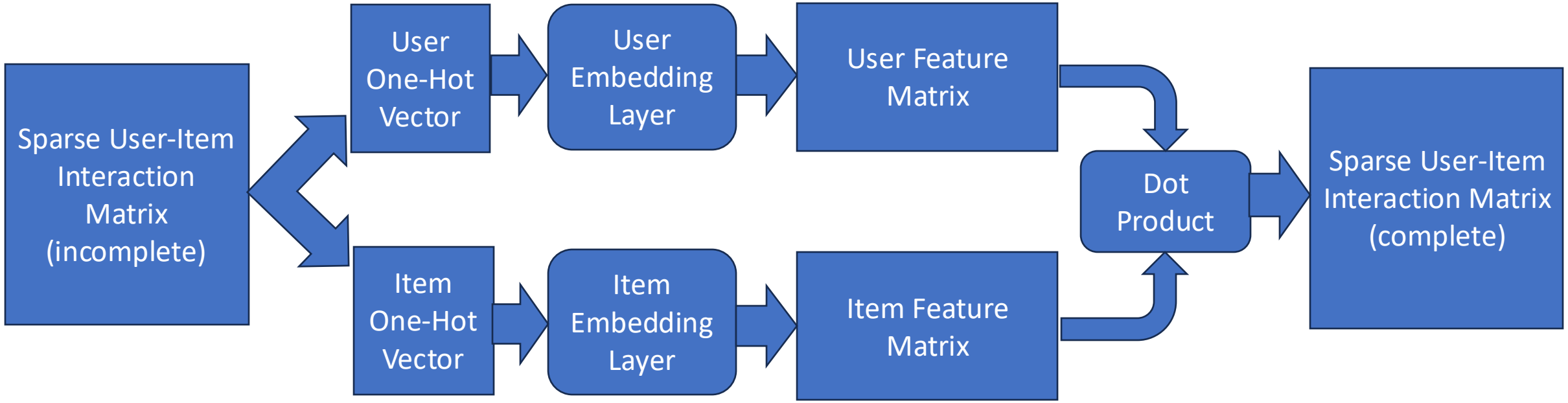
Flowchart of KNN-based recommender system



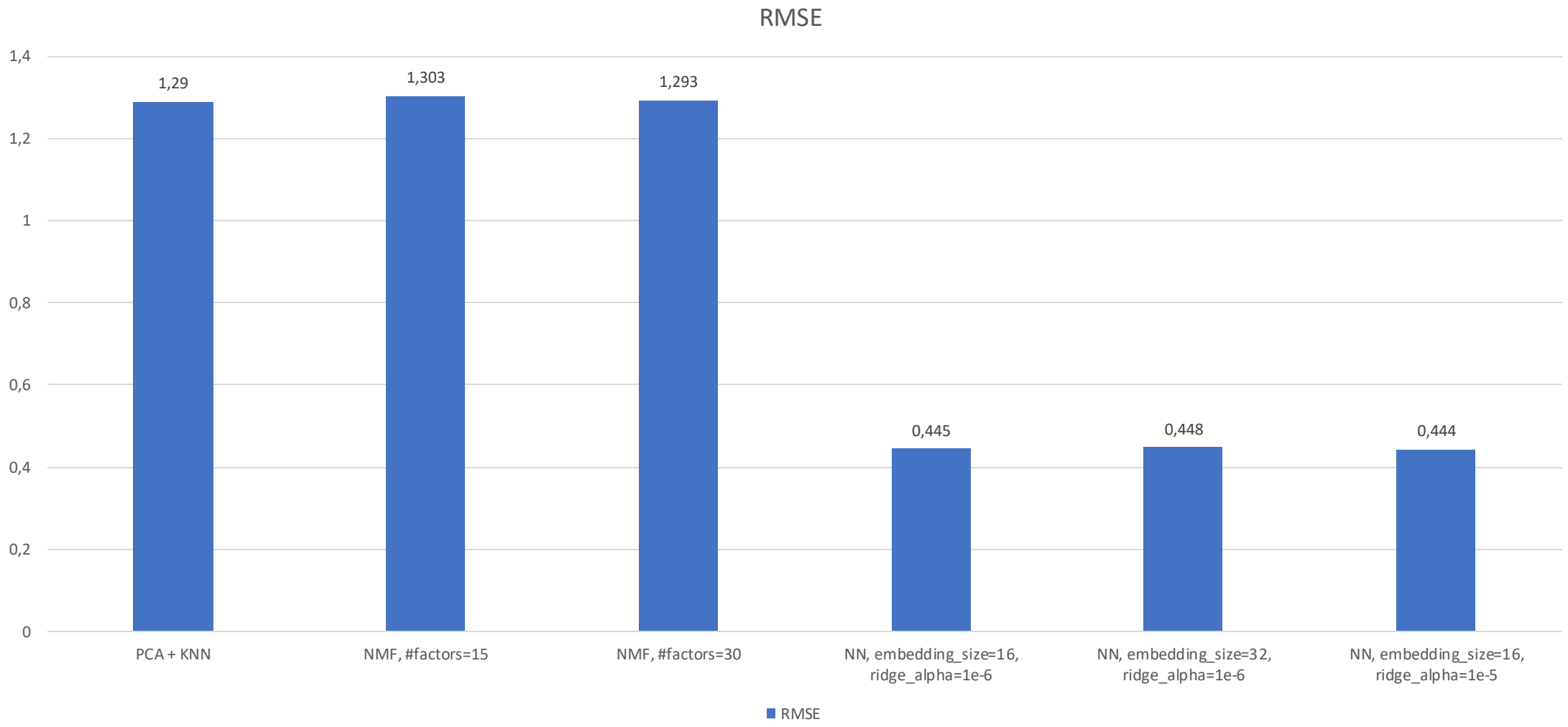
Flowchart of NMF based recommender system



Flowchart of Neural Network Embedding based recommender system



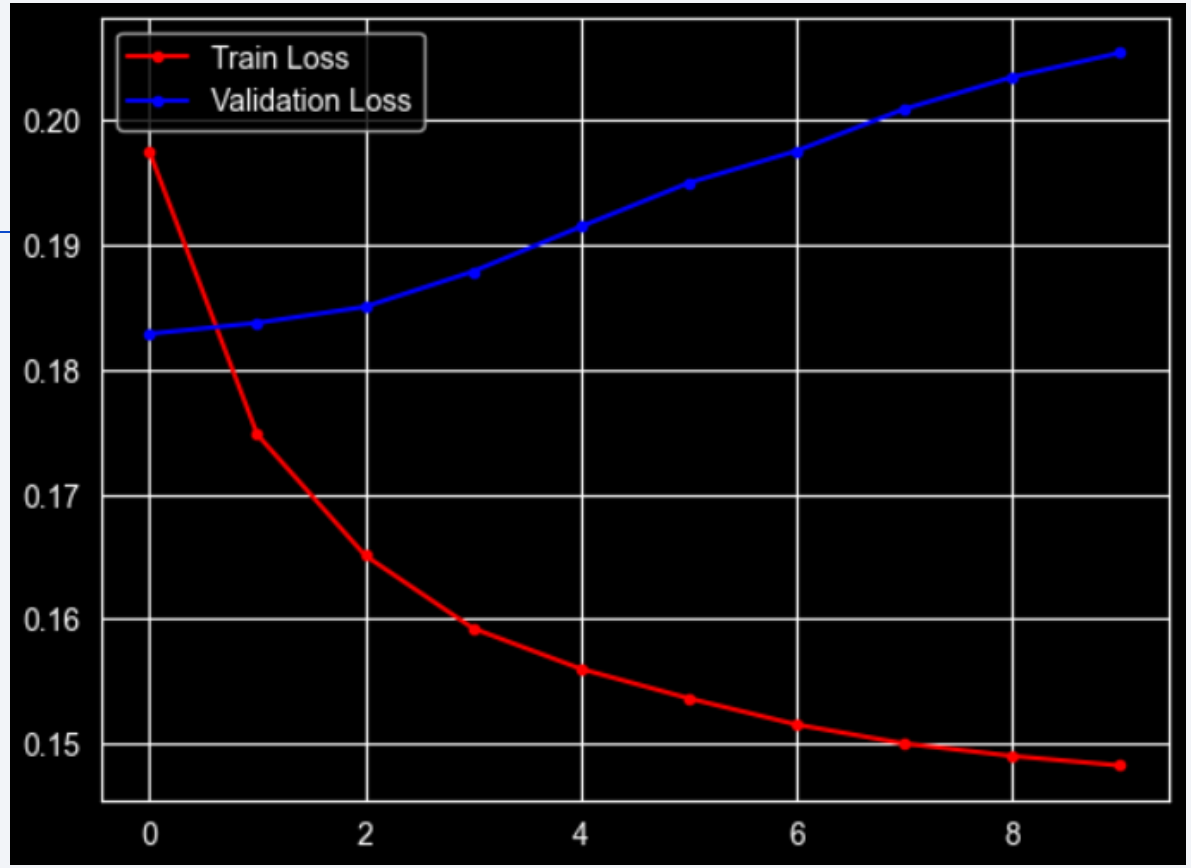
Compare the performance of collaborative-filtering models



Conclusions

- Clustering-based methods allow higher flexibility for an unsupervised recommender system
  - Variety of results depending on chosen clustering method, #clusters, threshold, etc.
- Content-based unsupervised recommender systems follow more of a deterministic approach
  - User profile-based system yielded a considerably higher #recommendations per user (20-60) compared with Course similarity-based system (1-3)
    - This generally happens because similarities among users are more frequent than those among items, also due to their relative amounts
    - On the other hand the User Profile-based method doesn't allow to recommend items of new genres to the target user
- NN Embedding-based resulted as the best-performing supervised recommender system
  - Overcome problems of data sparsity, cold start and scalability of standard collaborative filtering methods
  - But requires a large training dataset as well as careful hyperparameter tuning and regularization
- As part of future developments it's possible to investigate Regression-/Classification-based Recommenders integrating NN Embeddings as input features

Appendix

Best-performing model code & run history

```
embedding_size = 16
ridge_alpha = 1e-5
model = RecommenderNet(
    num_users, num_items, embedding_size, ridge_alpha
)
model.compile(
    optimizer=keras.optimizers.Adam(),
    loss=keras.losses.MeanSquaredError(),
    metrics=[tf.keras.metrics.RootMeanSquaredError()],
)
run_hist = model.fit(x_train, y_train, validation_data=(x_val, y_val), epochs=10, batch_size=64)
saved_model = model.save_weights('recommender_net_weights.weights.h5')

fig, ax = plt.subplots()
ax.plot(run_hist.history['loss'],'r', marker='.', label='Train Loss')
ax.plot(run_hist.history['val_loss'], 'b', marker='.', label='Validation Loss')
ax.legend()
```